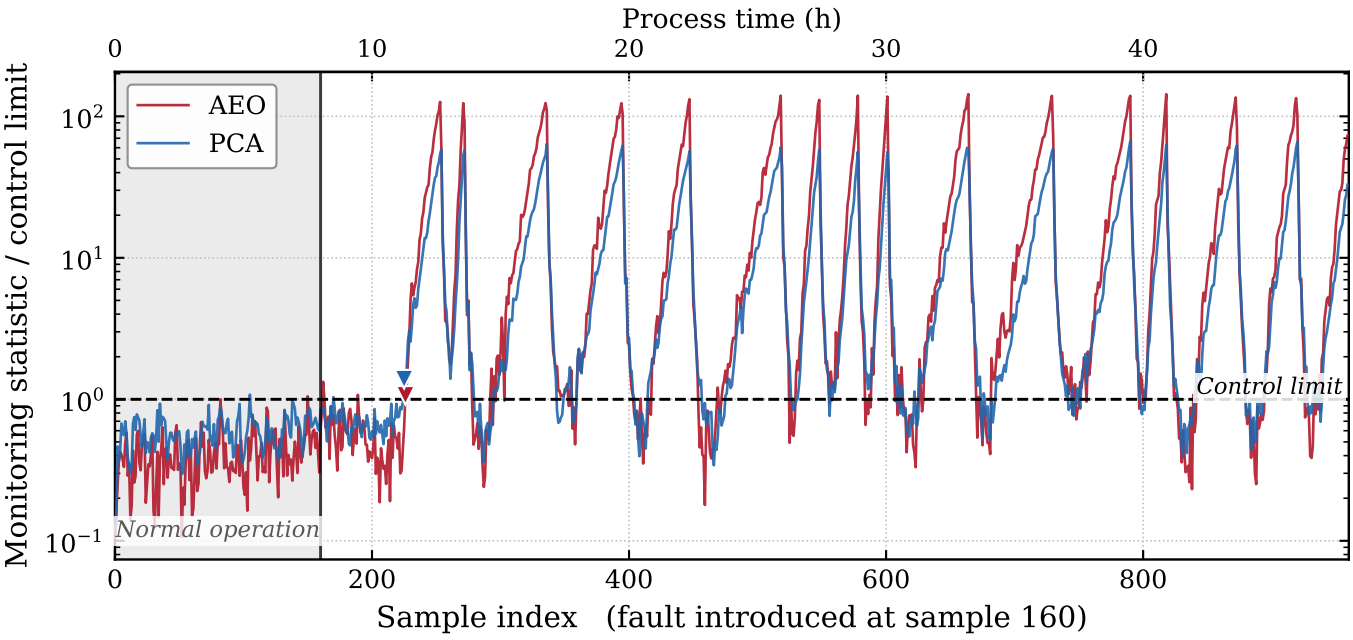
Fault Detection Performance for IDV(17)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	40.7 samples (122 min)	MDR 0.268	detected in 7/7 runs
PCA	delay	41.9 samples (126 min)	MDR 0.230	detected in 7/7 runs

AEO detects 1.1 samples (3 min) earlier; PCA missed-detection rate lower by 14%